



M.KUMARASAMY
COLLEGE OF ENGINEERING

NAAC Accredited Autonomous Institution

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ISO 9001:2015 Certified Institution

Thalavapalayam, Karur, Tamilnadu.



A Project work I Report

On

SYNTHESIS OF IRON OXIDE AND ITS WETTABILITY

Submitted in partial fulfilment of requirements for the award of the

Degree of

BACHELOR OF ENGINEERING

in

MECHANICAL ENGINEERING

Under the guidance of

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M.KUMARASAMY COLLEGE OF ENGINEERING
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BONAFIDE CERTIFICATE

Certified that this project report “**SYNTHESIS OF IRON OXIDE AND ITS WETTABILITY**” is the bonafide work of “**KARTHICK V (18BME4060), SABARISH P (18BME4112), SUBRAMANIAN PL (18BME4136)** ” who carried out the Minor project work during the academic year 2021-2022 under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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This project Report has been submitted for the Project work I viva voce Examination held on _____

EXAMINER

DECLARATION

We affirm that the Project report titled **SYSNTHESIS OF IRON OXIDE AND ITS WETTABILITY** being submitted in partial fulfillment for the award of **Bachelor of Engineering in Mechanical Engineering**, is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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SUBRAMANIAN PL

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Words are boundless to thank **Our Parents and Friends** for their constant encouragement to complete this project successfully.

Vision and Mission of the Institute

Vision

To emerge as a leader among the top institutions in the field of technical education.

Mission

1. Produce smart technocrats with empirical knowledge who can surmount the global challenges.
2. Create a diverse, fully-engaged, learner-centric campus environment to provide quality education to the students.
3. Maintain mutually beneficial partnerships with our alumni, industry and professional associations.

Vision and Mission of the Department

Vision

To create globally recognized competent Mechanical engineers to work in multicultural environment.

Mission

1. To impart quality education in the field of mechanical engineering and to enhance their skills, to pursue careers or enter into higher education in their area of interest.
2. To establish a learner-centric atmosphere along with state-of-the-art research facility.
3. To make collaboration with industries, distinguished research institution and to become a centre of excellence

Program Educational Objectives

The graduates of Mechanical Engineering will be able to

PEO1: Graduates of the program will accommodate insightful information of engineering principles necessary for the applications of engineering.

PEO2: Graduates of the program will acquire knowledge of recent trends in technology and solve problem in industry.

PEO3: Graduates of the program will have practical experience and interpersonal skills to work both in local and international environments.

PEO4: Graduates of the program will possess creative professionalism, understand their ethical responsibility and committed towards society.

PROGRAM OUTCOMES

Mechanical Engineering Students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

The Graduates will demonstrate the abilities

PSO 1: To comprehend, analyze, design and develop innovative products and provide solutions for the real-life problems.

PSO 2: To work collaboratively on multi-disciplinary areas and make quality projects.

PSO 3: To adopt modern tools, mathematical, scientific and engineering fundamentals required to solve industrial and societal problems.

1.ABSTRACT

- Iron oxides are common natural compounds and can also easily be synthesized in the laboratory. There are 16 iron oxides, including oxides, hydroxides, and oxide-hydroxides. These minerals are a result of aqueous reactions under various redox and pH conditions. They have the basic composition of Fe, O, and/or OH, but differ in the valency of iron and overall crystal structure.
- Enhancement in the quality of both organic and water soluble IO nanoparticles opens avenues of opportunities for development of IO nanoparticles based applications, for example:
 1. As contrast agents for Magnetic Resonance Imaging (MRI)
 2. As drug carriers for target specific drug delivery
 3. As gene carriers for gene therapy

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2.LIST OF FIGURES

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3.LIST OF ABBREVIATIONS

- 1. FeSo₄-Ferrous Sulphate**
- 2. NaOH-Sodium Hyroxide**
- 3. FeO₂-Iron Oxide**

4.INTRODUCTION

Several methods have been developed to produce metal nanoparticles. Two synthesis approaches have been identified that is top-down and bottom-up approach. Top-down methods comprise of milling, lithography, and repeated quenching. This approach does not have good control of the particle size and structure. Bottom-up method is the approach that is mostly used by scientists in the synthesis of nanoparticles as it involves building up a material from bottom: atom-by-atom, molecule-by-molecule, and cluster-by-cluster. Several chemical routes have been identified to synthesize the colloidal metal nanoparticles from different precursors using chemical reductants in solvents (aqueous and nonaqueous).

5.DIFFERENT METHODS

- Chemical method
 - 1.Sol-gel
 - 2.Chemical vapour deposition
 - 3.Colloidal method
 - 4.Spray pyrolysis
- Biological method
 - 1.Green synthesis
- Physical method
 - 1.Mechanical method
 - 2.Vapour method

6.SOL-GEL METHOD

- The sol–gel process is a wet chemical technique also known as chemical solution deposition, and involves several steps, in the following chronological order: hydrolysis and polycondensation, gelation, aging, drying, densification, and crystallization.
- One advantage of the sol–gel technique is that it is an easy and very cheap process to prepare metal oxides and allows control over the doping process or addition of transition metals, as compared to other preparation techniques.

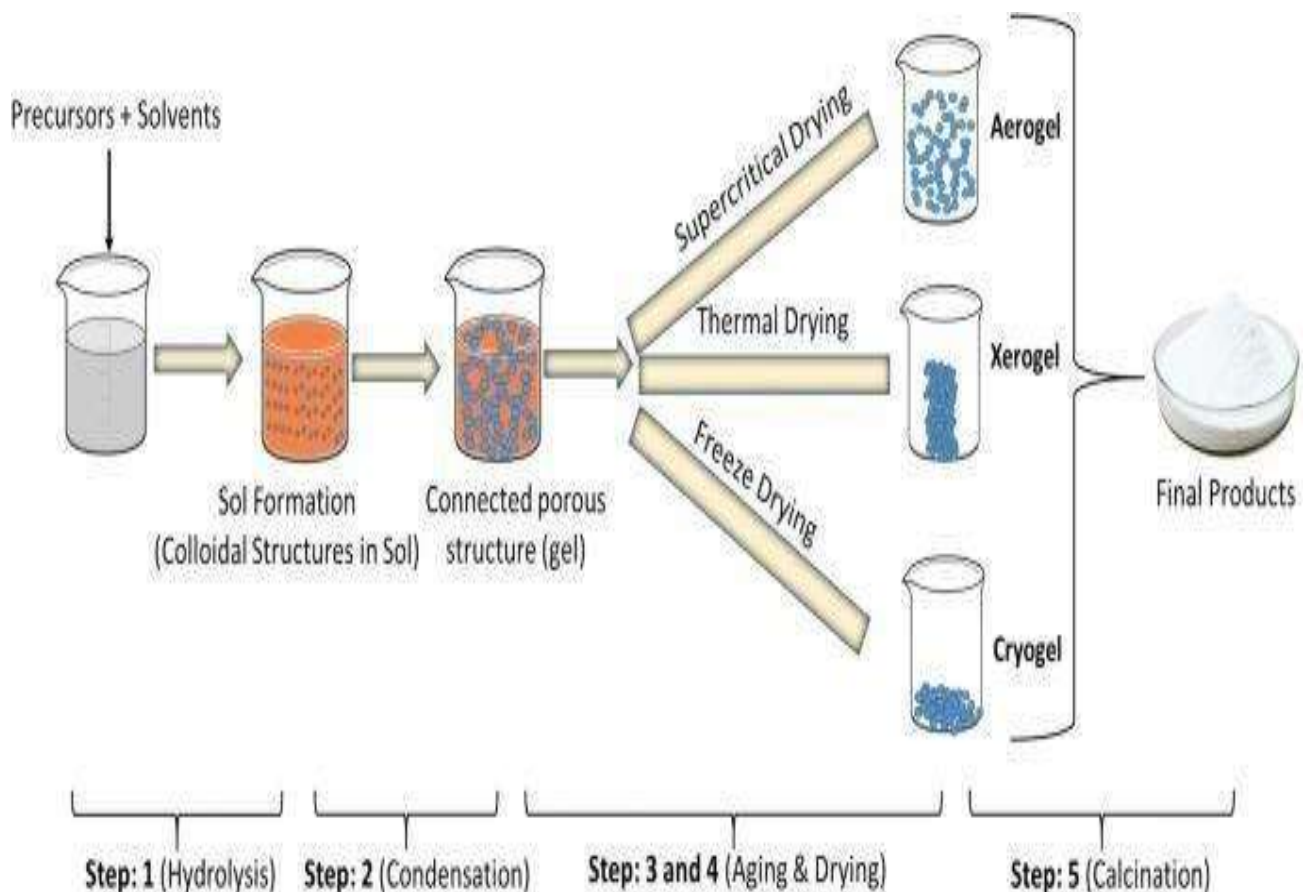


Figure 1-Sol-gel method

7. STIRRING

- 4 gram of FeSO_4 is diluted with 100ml of distilled water
- 40 ml of solvent(NaOH) is added to the FeSO_4 solution
- The solution is stirred for 3hour until the solvent get mixed with the FeSO_4 solution



Figure 2- FeSO_4 solution

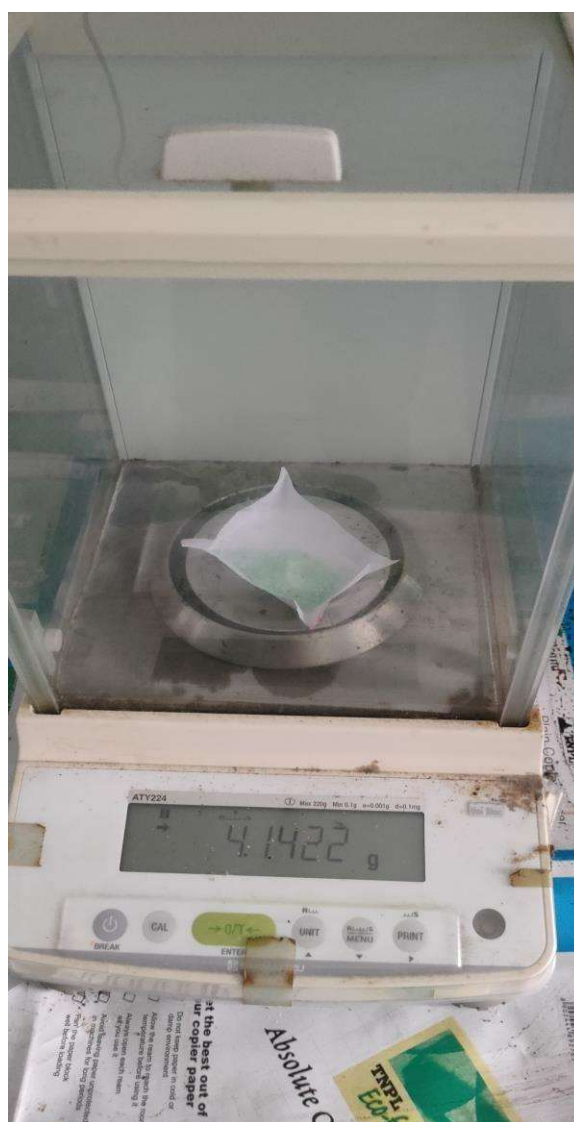


Figure 3-Weight of the salt

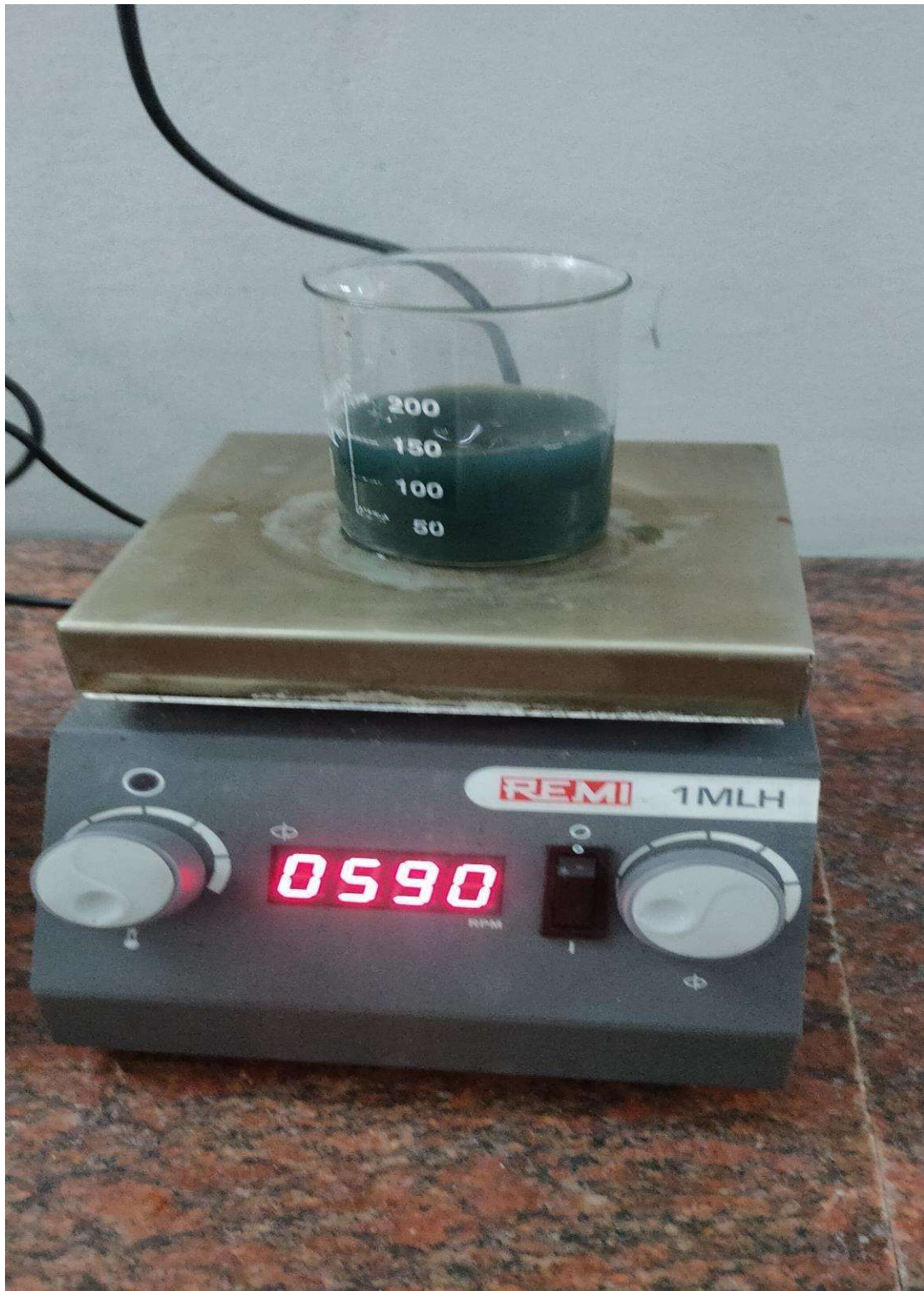


Figure 4- $\text{FeSO}_4 + \text{NaOH}$ solution

- After heating the solution, we get the IRON OXIDE particle.

8.BALL MILLING

- A ball mill is a device used to grind and blend materials for use in mineral dressing processes, paints, pyrotechnics, ceramics and selective laser sintering. It is a physical method of synthesis of nanoparticles and is an example of top down approach of producing nanomaterials.





Figure 5-Ball Milling Process

9.CONTACT ANGLE

- Contact angle, θ (theta), is a quantitative measure of wetting of a solid by a liquid. The contact angle is geometrically defined as the angle formed by a liquid at the three-phase boundary where a liquid, gas, and solid intersect.
- We measure the contact angle for three different amount of IRON OXIDE nanofluid.
 1. 250 ml of 0.009g iron oxide mixed nanofluid.
 2. 250 ml of 0.09g iron oxide mixed nanofluid.
 3. 250 ml of 0.9g iron oxide mixed nanofluid.

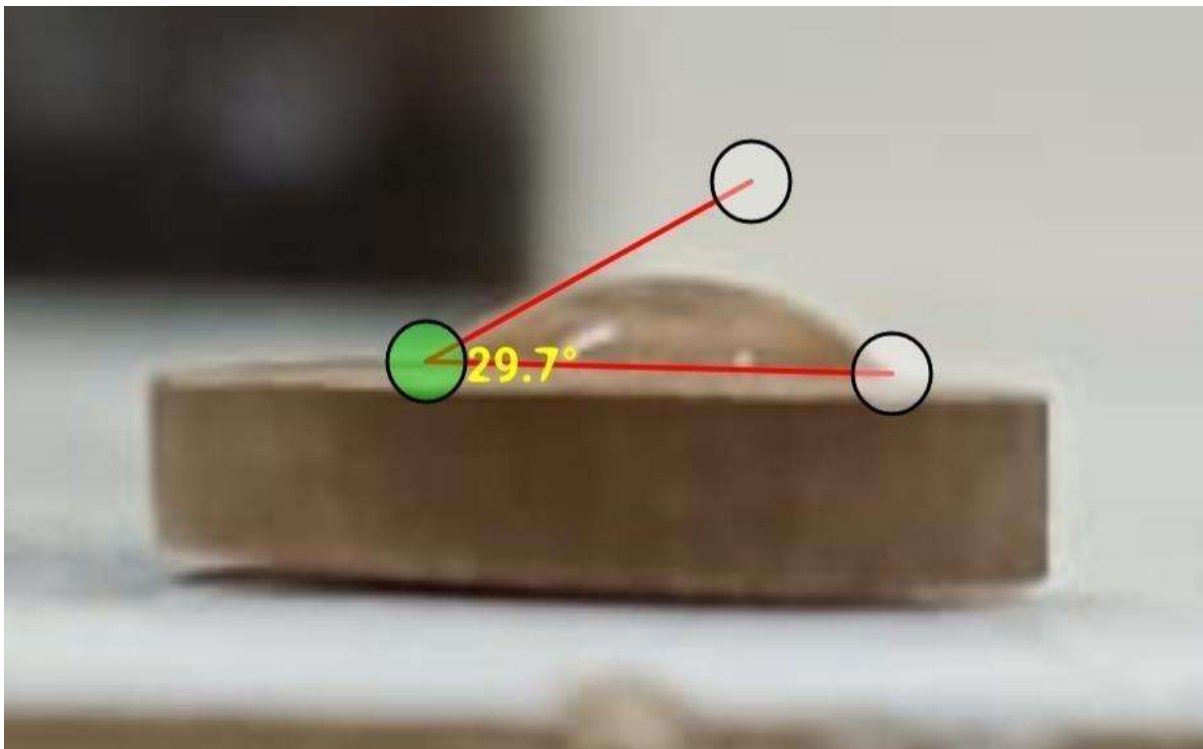


Figure 6-0.009g IRON OXIDE mixed nanofluid

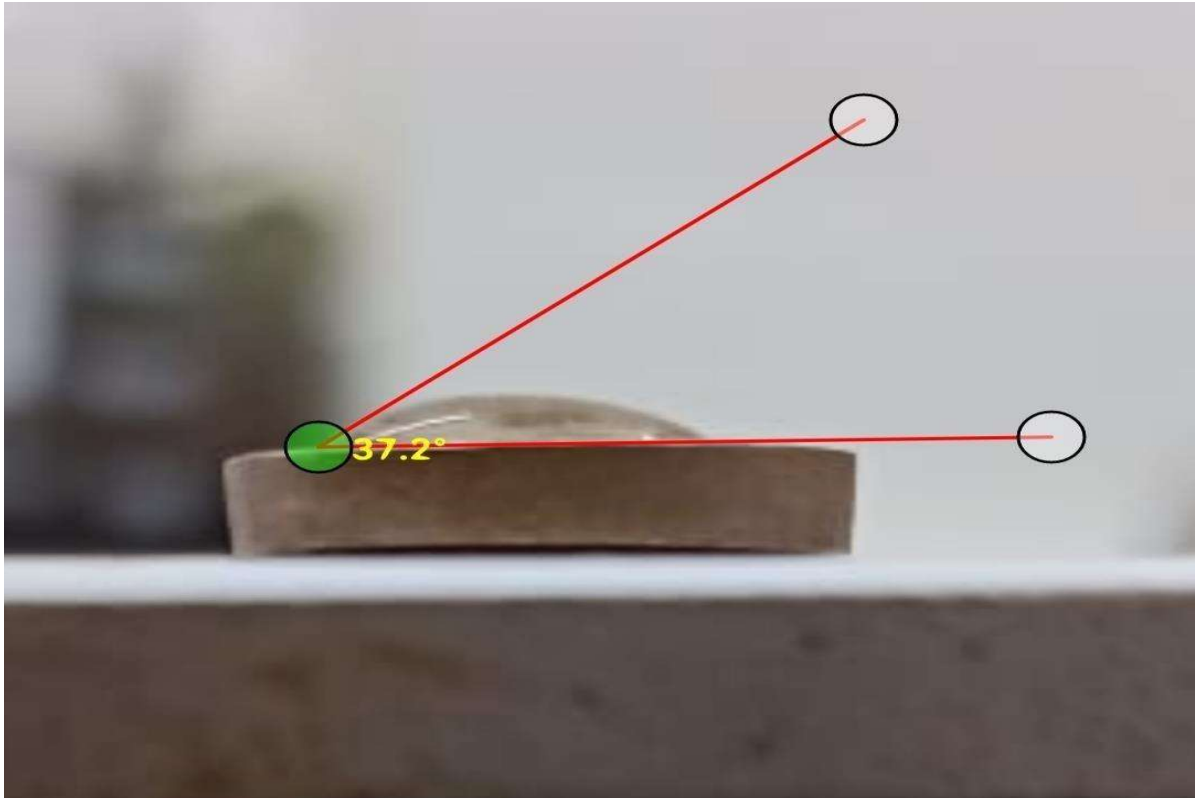


Figure 7-0.09g IRON OXIDE mixed nanofluid

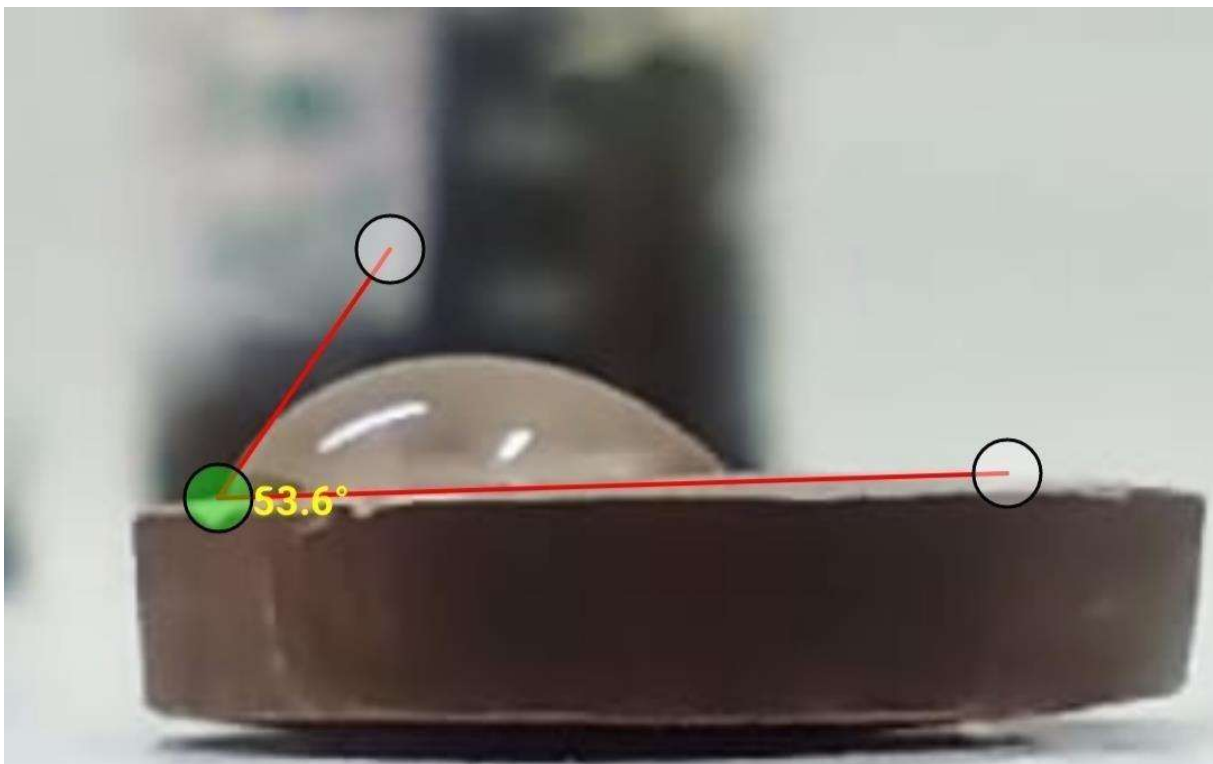


Figure 8-0.9g IRON OXIDE mixed nanofluid

10.RESULT

- Different contact angle is obtained for the three different amount of iron oxide mixed nanofluid.

Amount of iron oxide mixed	Contact angle
0.009	29.7 ⁰
0.09	37.2 ⁰
0.9	53.6 ⁰

11.REFERENCE

- Synthesis and characterization of Ni doped SnO₂ nanoparticles by sol-gel method for novel applications,P. Srinivasa Subbarao , Y. Aparna , Kanchana Latha Chitturi.
- Preparation, characterization and antifungal activity of iron oxide nanoparticles, Shazia parveen, Abdul hamid wani.